Homework for Chapter 7

1. Study with the data from any source (reference books, websites, AI apps), and compare the transverse and longitudinal eigen-frequencies of optical phonons of a series of semiconductors of ZnO, ZnS, ZnSe, and ZnTe, what do you conclude?

2. Compare the longitudinal-transverse splitting at $k = 0$ for Ge and the corresponding III–V, II–VI and I–VII compounds on the same row of the periodic table of the elements. What do you find and qualitatively explain your findings.